

# Quantum Machine Learning: An Overview

There are quite a few open source libraries and frameworks available for quantum machine learning, and most of them are compatible with Python programming.



Quantum computing is based on the concept of superposition, in which 0 and 1 can both occur at the same time — something not possible in classical computers. Its high-performance supercomputing capabilities can solve problems in multiple domains including healthcare, finance, cyber security and education.

Realising the importance of quantum computing, the National Quantum Mission (NQM) of India is focusing on quantum communications, quantum sensing, quantum metrology, quantum materials, and quantum devices. It has set up four Thematic Hubs (T-Hubs) at leading institutions in the country for this purpose.

## Quantum machine learning: Application areas

Quantum machine learning (QML) replaces classical machine learning applications with quantum implementations, helping optimise mammoth computations in fields like fintech (integration of finance and technology), blockchain, cryptography, logistics, and supply chain management. Here are some examples of how QML helps in different domains.

- **Fintech and blockchain:** Transaction validation, banking token generation, smart contracts, consensus optimisation, security auditing

- **Finance:** Portfolio optimisation, fraud detection, asset allocation, risk analysis, option pricing
- **Supply chain management:** Demand forecasting, inventory optimisation, supplier evaluation, shipment tracking, route planning
- **Cryptography:** Secured key generation, encryption analysis,

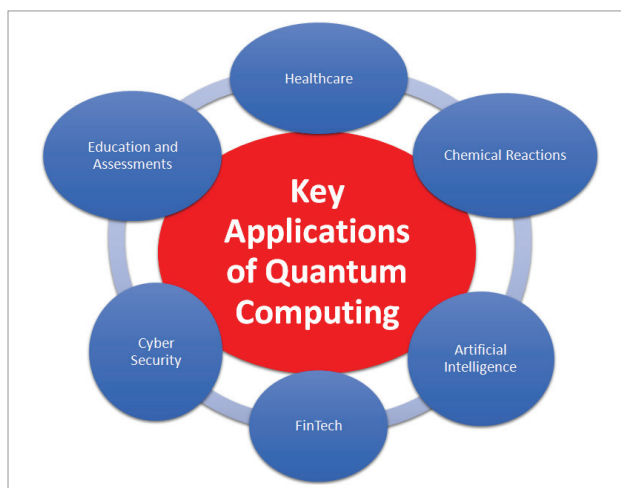


Figure 1: Key applications of quantum computing